Technical Support Fundamentals

1. System Health Audit

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- **Platform:** Linux (ISH Environment) /CLI
- **DATE:** February 2026

➢ In Course 1 of the Google IT Support Professional Certificate ,I learned how hardware components interact with the Operating System.For this project, I developed a shell script to automate the the retrieval of system specifications, a common task for IT Support Specialist diagnosing performance issues.

- **Hardware Components Monitored**

Components	Command Used	Purpose in Troubleshooting
Central Processing Unit (CPU)	uptime	Check system load and how long since last reboot.
Random Access Memory (RAM)	free -m	Identifies if the system is running out of active memory
Storage (Disk)	df -h	Checks if the disk is full, which can stop OS updates

- **<u>The Script Code</u>**

```
#!/bin/sh

# System Health Check Script

uptime

free -m

df -h
```

➢ As part of a proactive maintenance strategy, I developed this System Health Audit utility. In an IT Support role, quickly identifying hardware bottlenecks such as low disk space high memory usage is essential for reducing system downtime.The following screenshot demonstrates the script’s ability pull live data from the Linux kernel to provide an immediate diagnostic snapshot.

```
als
localhost:~/Portfolio# ls
Technical_Support_Fundamentals
localhost:~/Portfolio# vim hardware_audit.sh
localhost:~/Portfolio# ls
Technical_Support_Fundamentals
hardware_audit.sh
localhost:~/Portfolio# chmod +x hardware_audit.sh
localhost:~/Portfolio# ./hardware_audit.sh
./hardware_audit.sh: line 1: o: not found
SYSTEM HEALTH REPORT: COURSE 1
DATE: Fri Feb 27 07:24:33 UTC 2026
==========================================
1. CPU & UPTIME:
 07:24:33 up 1 day,  7:07,  load average: 1.12, 1.58, 1.
77

2. RAM/MEMORY USAGE:
              total        used        free      shared
 buff/cache   available
Mem:              0           0           0           0
          0           0
-/+ buffers/cache:            0           0
Swap:             0           0           0

3. DISK STORAGE:
Filesystem                Size      Used Available Use%
Mounted on
                        119.1G    115.6G      3.5G  97%
/
==========================================
localhost:~/Portfolio#
```

- The audit revealed that Disk Storage is at 97% capacity(only 3.5GB available.

- The system is at risk of instability due to low disk space.

- Temporary log files or cached data are likely occupying the drive.

- Search for large file and perform a system cleanup to prevent crashes.

- Re-run hardware_audit.sh to confirm the **Use%** has decreased.

- **Standard Operating Procedure:The Troubleshooting Framework**

- Gather information identify symptoms, and determine if anything has changed.

- List obvious causes first and research internal/ external symptoms.

- Confirm your theory or move on to a new one if it fails.

- Develop a plan to fix the issue and carry it out.

- Ensure the issue is truly resolved and implement preventive measures.

- Record the problem and the fix for future reference.

2. User Access & Security Audit

- Security is among the most important IT Support areas to be aware of. While conducting this audit I made certain to add an evaluation for User Access because in any business setting, it is the responsibility of the technician to verify only authorized personnel have access to a specific workstation.

- If an audit reveals an unknown user account or a ‘Guest’ account with high level permissions, the technician must identify this as a security breach and follow’s company Insolation and Escalation.

```
localhost:~/Portfolio/Technical_Support_Fundamentals# ./
hardware_audit.sh

4. SECURITY & USER ACCESS:
Current User: root
Users currently logged in:
./hardware_audit.sh: line 4: who: not found
Last 3 Login attempts:
./hardware_audit.sh: line 6: last: not found
./hardware_audit.sh: line 7: o: not found
SYSTEM HEALTH REPORT: COURSE 1
DATE: Thu Mar  5 19:41:07 UTC 2026
==========================================
1. CPU & UPTIME:
 19:41:07 up 15:44,  load average: 1.43, 1.71, 1.91

2. RAM/MEMORY USAGE:
              total        used        free      shared
 buff/cache   available
Mem:              0           0           0           0
          0           0
-/+ buffers/cache:            0           0
Swap:             0           0           0

3. DISK STORAGE:
Filesystem                Size      Used Available Use%
Mounted on
                        119.1G    116.8G      2.3G  98%
/
==========================================
localhost:~/Portfolio/Technical_Support_Fundamentals#
```

- By integrating a User Access Audit into the hardware diagnostic, I have created a holistic ‘System Health Check’. This allows a technician to address performance issues and security vulnerabilities simultaneously, ensuring the workstation is both efficient and secure.